

PLANETVORTEX

PLANETVORTEX — Lemma D — the Kepler register and the mass correction

The theorem edition of the planetary gravity-geometry bench — mini-repository planetvortex/ of the TRIVORTEX framework, version 1.0.0. This dossier binds one statement to its registered protocol; the full science lives in the big monograph (docs/monograph/).

Statement

For a planet of mass ratio $m_i/M_\odot$ on the committed orbit (a_i, e_i) the two-body period is $T_i = 2\pi \sqrt{a_i^3 / (GM_\odot(1+m_i/M_\odot))}$. The period measured from interpolated perihelion passages of the RK4 two-body flow matches T_i to 6.2×10^{-13} relative across all eight planets; the uncorrected formula $2\pi \sqrt{a_i^3 / GM_\odot}$ misses by the exact signature $\sqrt{1+m_i/M_\odot} - 1 = m_i/2M_\odot + O(m_i^2)$, up to 4.77×10^{-4} for Jupiter.

Proof

The two-body relative flow is exactly Keplerian, so the measured period converges to T_i at the integrator's phase accuracy; the parabolic interpolation of the perihelion minima cancels its systematic bias between successive minima to $(ndt)^4$. The uncorrected ratio is

$T_{uncorr}/T_i = \sqrt{\mu_i/4\pi^2} = \sqrt{1+m_i/M_\odot}$ exactly, whence the signature. The recorded separation of the two registers is a factor 7.7×10^8 ; the committed fact-sheet periods are recorded as a provenance diagnostic at 4.8×10^{-4} (Neptune worst) — rounded and epoch-mixed. ▢

Protocol binding

item	value
stage	P2
recorded registers	6.2×10^{-13}
protocol file	results/protocols/ (default preset)
test guard	tests/ re-derives every preset-independent register

Cross-references

- the parent framework: Theorem 3.1 and the three-vortex anchor;

- the sibling bench polyvortex: the W2/W4 registers at $N=7$ — the same heptagon, cross-pinned bit-for-bit;
- the big monograph: Sections 3 (Theorem A), 6 (Theorem C), 7 (Theorem B), 4 (Lemma D), 8 (Lemma E);
- the formalization queue: roadmap item V4 (Coq / Lean 4).